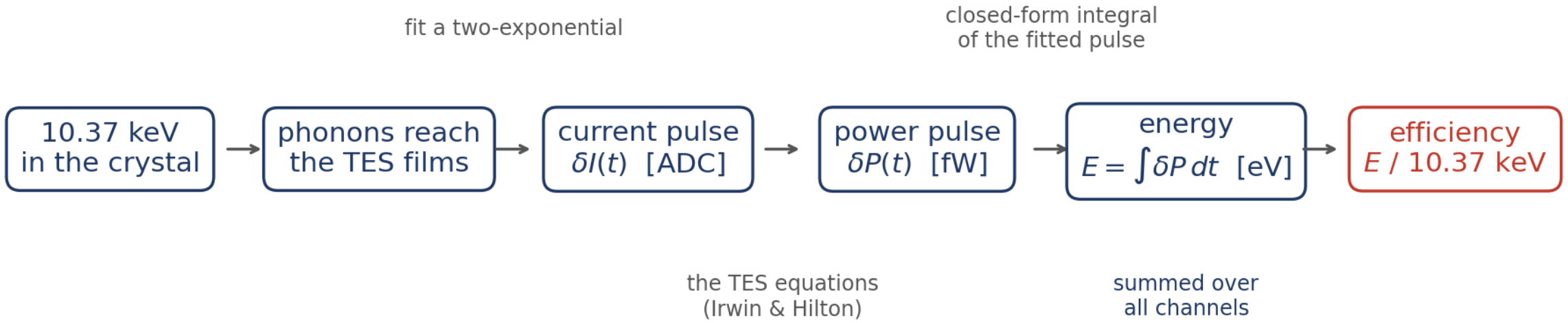
Phonon collection efficiency of Z7

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github.com/ZhihengLi0/urop → snolab_run4_phonon_pulse_analysis/collection_efficiency/

The Main Idea



For detectors operated at 0 V (no NTL effect):

Collection efficiency = energy absorbed by the TESs, summed over all channels ÷ the 10.37 keV of the event

Sample: Z7, the K-line events of the Ge activation run, from SNOLAB R4 (1442 events, 17 series, all 11 channels).

The formula, and the two numbers it needs

Method 1 of the CDMS collection-efficiency note

$$\delta P(t) \;=\; I_0\,(R_L - R_0)\,\delta I(t) \;+\; 2\,R_L\,(\delta I(t))^2$$

linear term quadratic term

$$E \;=\; \int \delta P(t)\,dt \;=\; I_0(R_L - R_0)\,A(\tau_f - \tau_r) \;+\; 2R_L\,A^2\left(\frac{\tau_f + \tau_r}{2} - \frac{2\tau_f\tau_r}{\tau_f + \tau_r}\right)$$

for the fitted pulse $\delta I(t) = A\,[\,e^{-t/\tau_f} - e^{-t/\tau_r}\,]$ integrated from $-\infty$ to $+\infty$: zero before it starts, decays to zero after

Z7 PBS1: $I_0 = -18.07\ \mu\text{A}$, $R_0 = 44.16\ \text{m}\Omega$, $R_L = R_p + R_{sh} = 14.24\ \text{m}\Omega + 5.00\ \text{m}\Omega = 19.24\ \text{m}\Omega$

linear coefficient: $I_0(R_L - R_0) = (-18.07\ \mu\text{A}) \times (19.24\ \text{m}\Omega - 44.16\ \text{m}\Omega) = 0.45\ \mu\text{V}$

quadratic coefficient: $2R_L = 2 \times 19.24\ \text{m}\Omega = 38.5\ \text{m}\Omega = 0.0385\ \Omega$

δP = the change of Joule heating when the current moves by δI: a linear term and a small quadratic term.

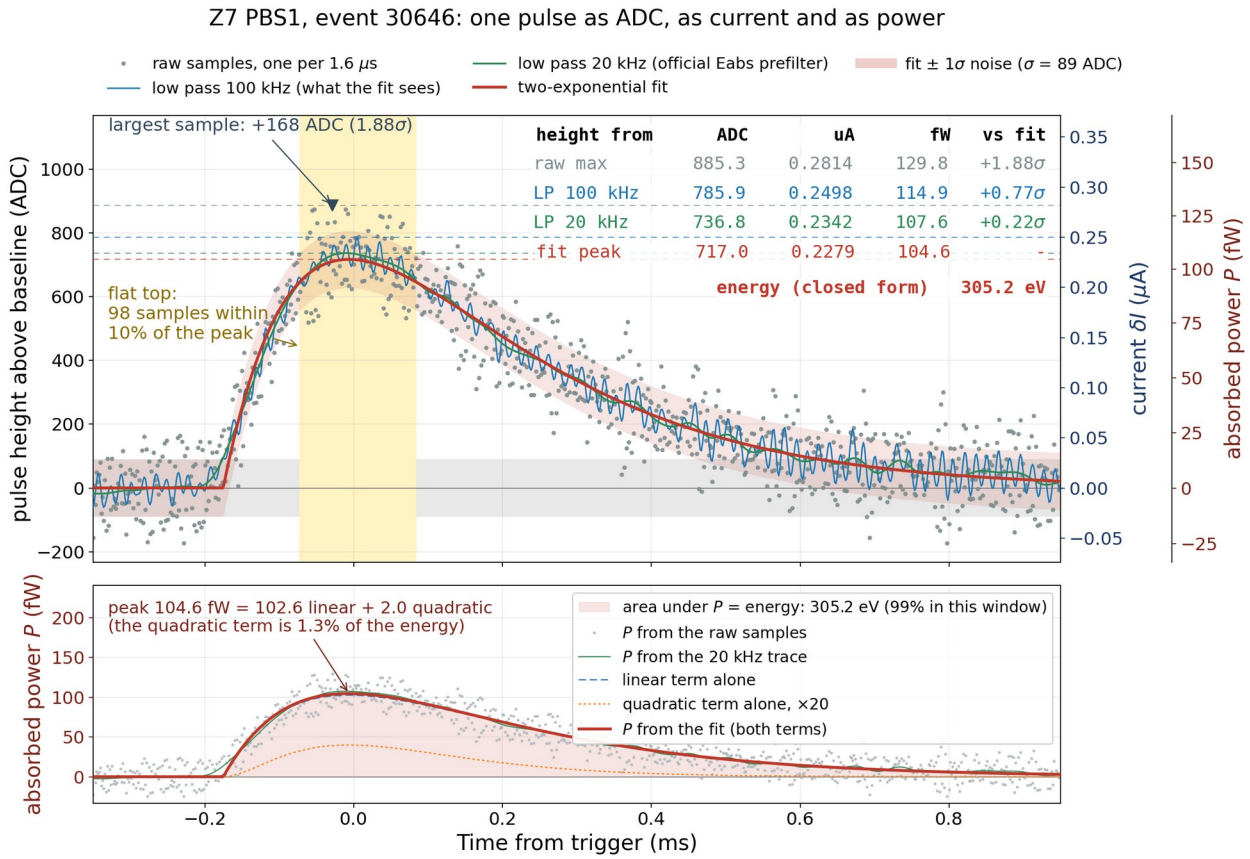
I_0: TES current at the bias point · R_0: TES resistance at the bias point · R_L = R_p + R_sh: load resistance (R_p parasitic, R_sh shunt) · δI: change in current, from the trace

The note also gives Method 2: $\delta P = I_0(R_L - R_0)\cdot\delta I - R_L\cdot(\delta I)^2$ in our sign convention — same linear term, only the quadratic term differs, 1.7 % less energy (PBS1, median of 15 events).

Source: Antoine Rehberg, CDMS wiki, Collection Efficiency Analysis — confluence.slac.stanford.edu/spaces/CDMS/pages/542577254

Example: one pulse

Z7 PBS1, event 30646: **the same trace as ADC counts, as current, as power**



baseline = mean of bins 93-15758 = 28667.7 ADC, subtracted | 1 ADC = 0.3179 nA

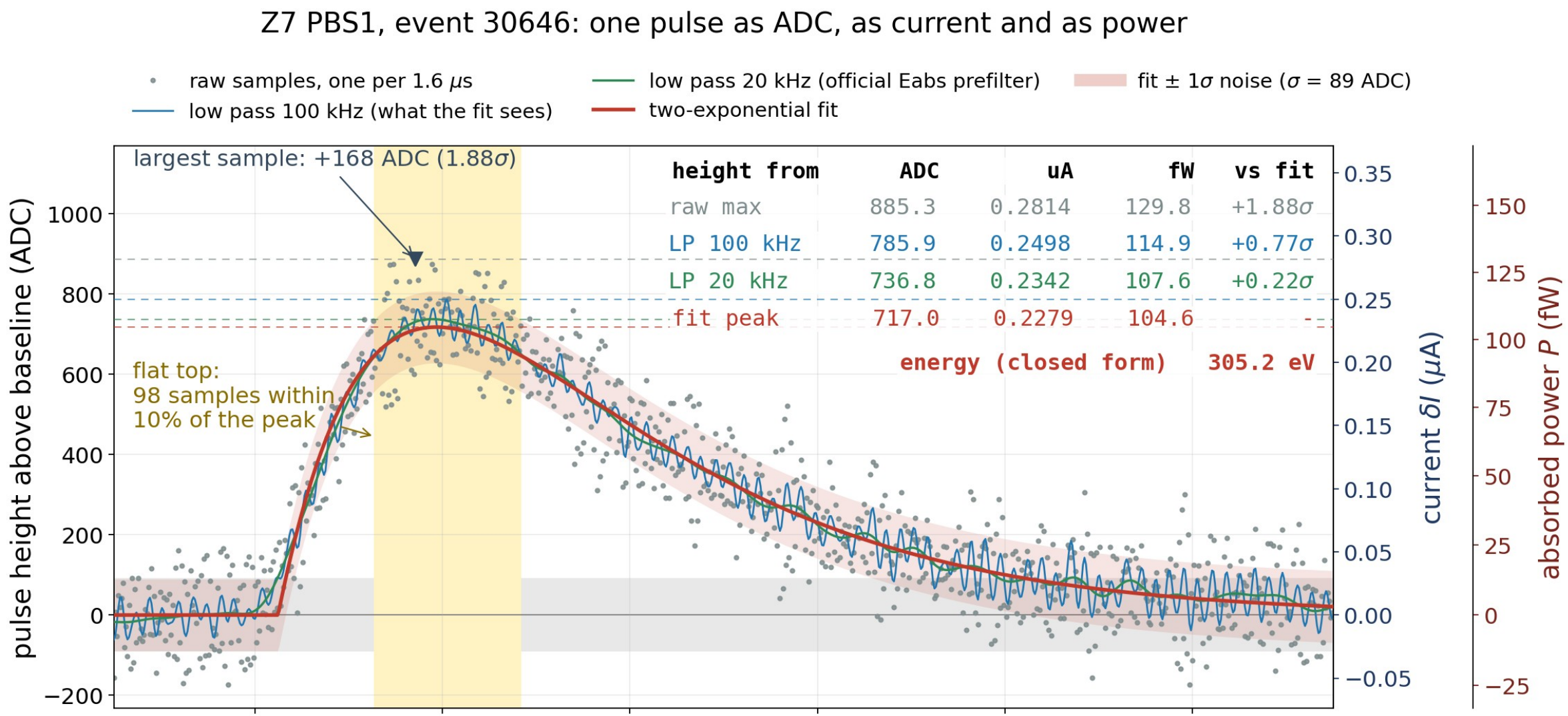
$\delta P = I_0(R_L - R_0)\,\delta I + c_2 R_L (\delta I)^2$ (method1, $c_2 = 2$), with I_0 = -18.069 μA, R_0 = 44.159 mΩ, R_L = 19.240 mΩ

linear coefficient: $I_0(R_L - R_0)$ = (-18.069 μA) × (19.240 mΩ − 44.159 mΩ) = 0.450 μV

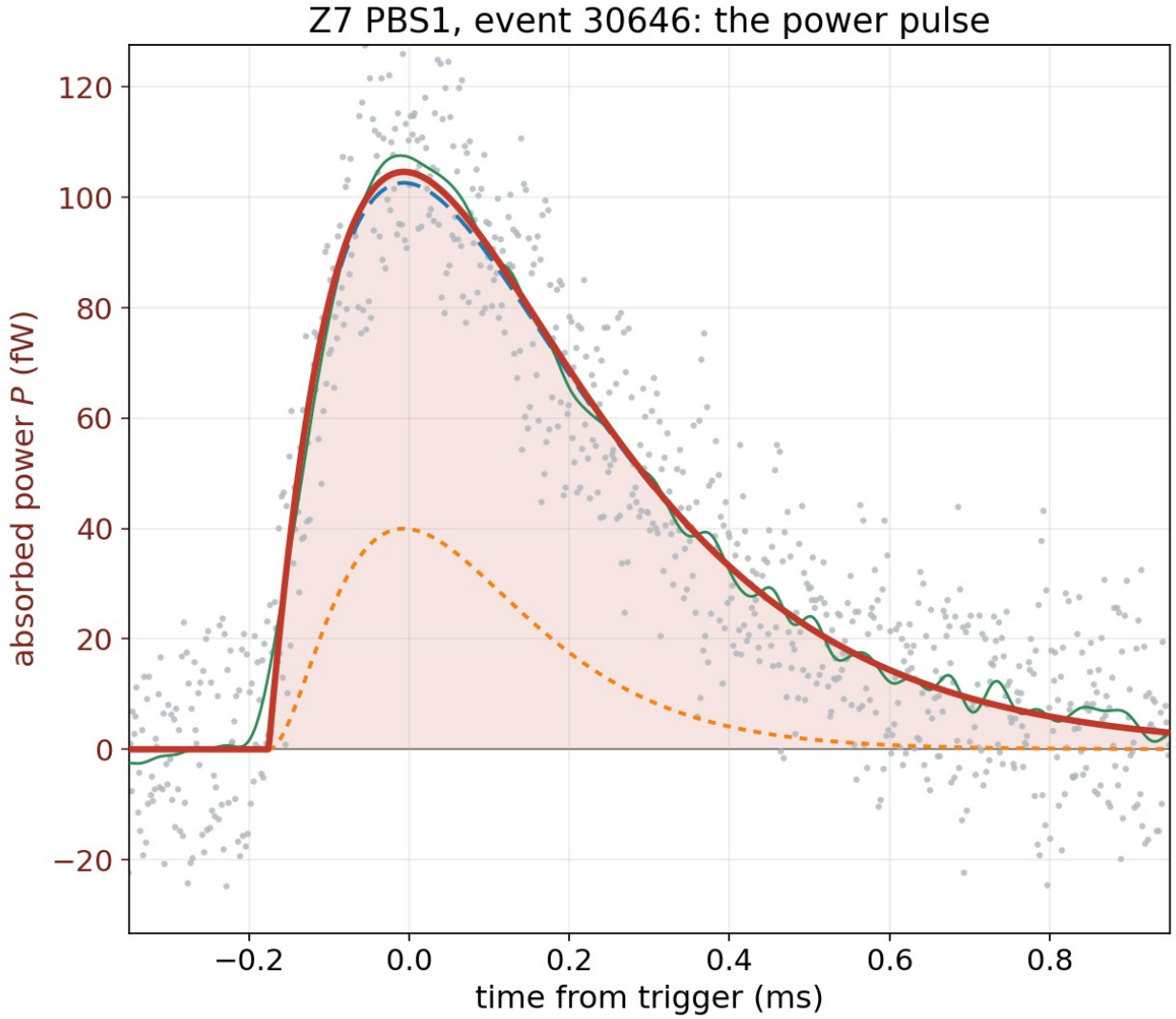
quadratic coefficient: $c_2 R_L$ = 2 × 19.240 mΩ = 38.48 mΩ = 0.0385 Ω

The two-exponential is fitted to the 100 kHz low-passed trace (blue), not to the raw samples; the raw samples are only drawn.

The height: take it from the fit, not from the largest sample



The power pulse, and its area is the energy



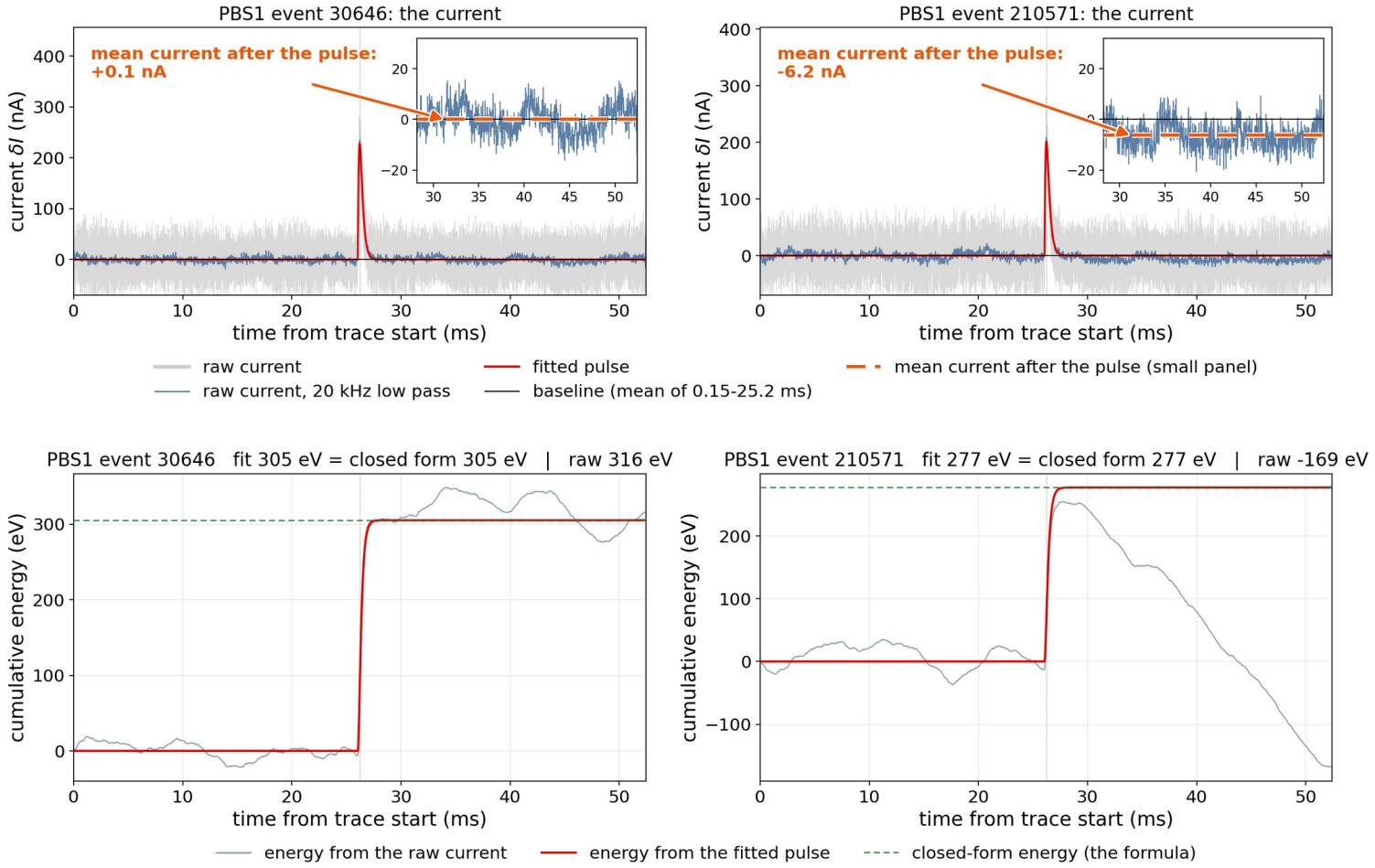
what is in the figure

- power from the fit, both terms
- the linear term alone
- the quadratic term alone, ×20
- from the raw samples, for comparison
- from the 20 kHz trace, for comparison
- area = the energy absorbed here

the quadratic term is 1.3 % of the energy,

so the power pulse has the shape of the cu

Why we integrate the fit, not the raw data

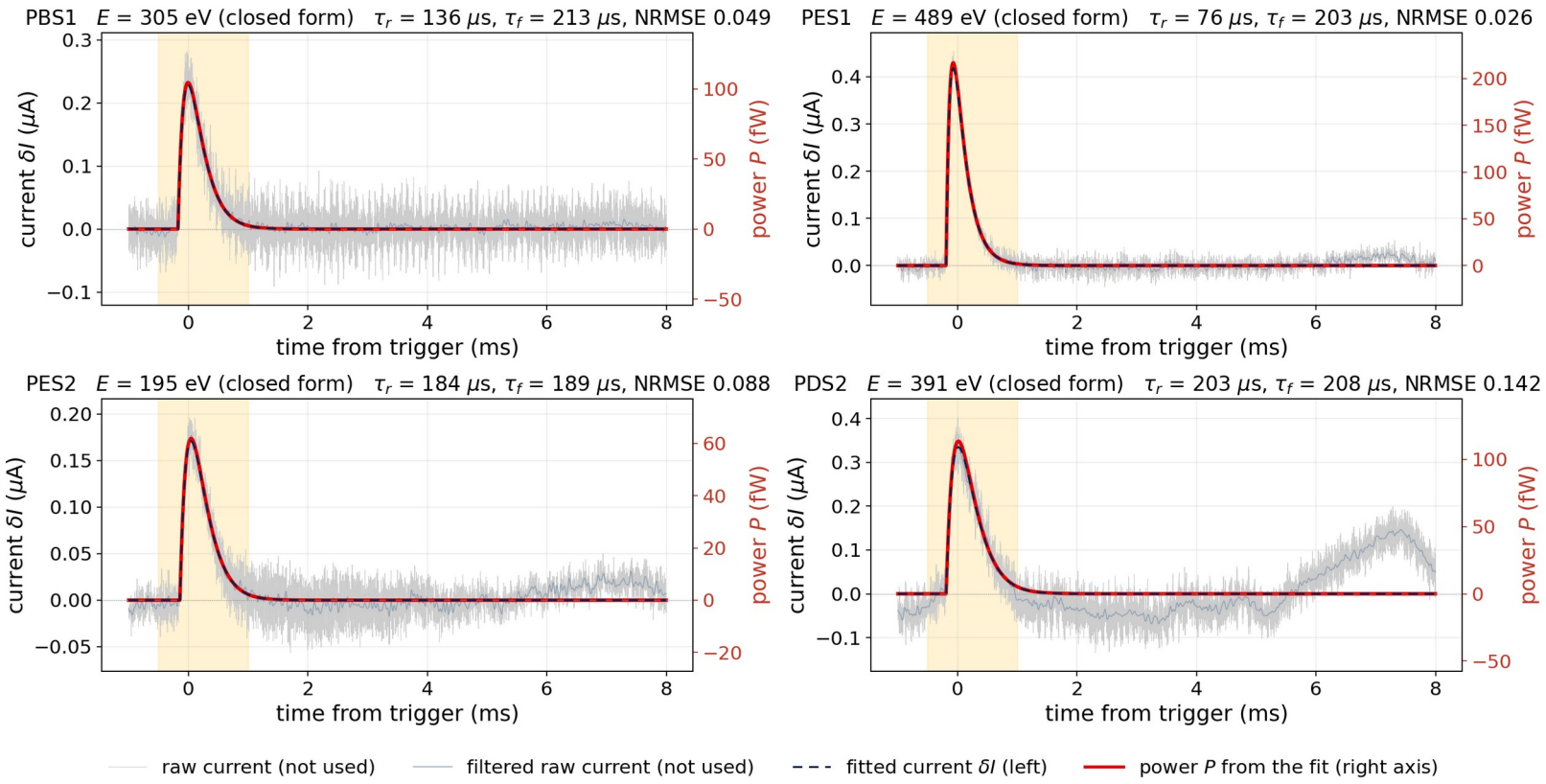


Top: the raw current of the two events; the small panel zooms in on the baseline after the pulse. Bottom: the energy accumulated from the raw current (grey) and from the fit (red).

Right: after the pulse the current sits 6 nA below its baseline — hidden in 28 nA of noise, yet over 24 ms it is −420 eV: raw −169 eV, fit 277 eV.

The same event in its other channels: they add up to 33 %

Z7 event 30646; four of the eleven channels



energy per channel, event 30646

side 1	**2062 eV (60 %)**
PAS1	303 eV
PBS1	305 eV
PCS1	227 eV
PDS1	378 eV
PES1	489 eV
PFS1	359 eV
side 2	**1357 eV (40 %)**
PAS2	267 eV
PBS2	261 eV
PCS2	243 eV
PDS2	391 eV
PES2	195 eV

total 3419 eV = 33.0 % of 10.37 keV

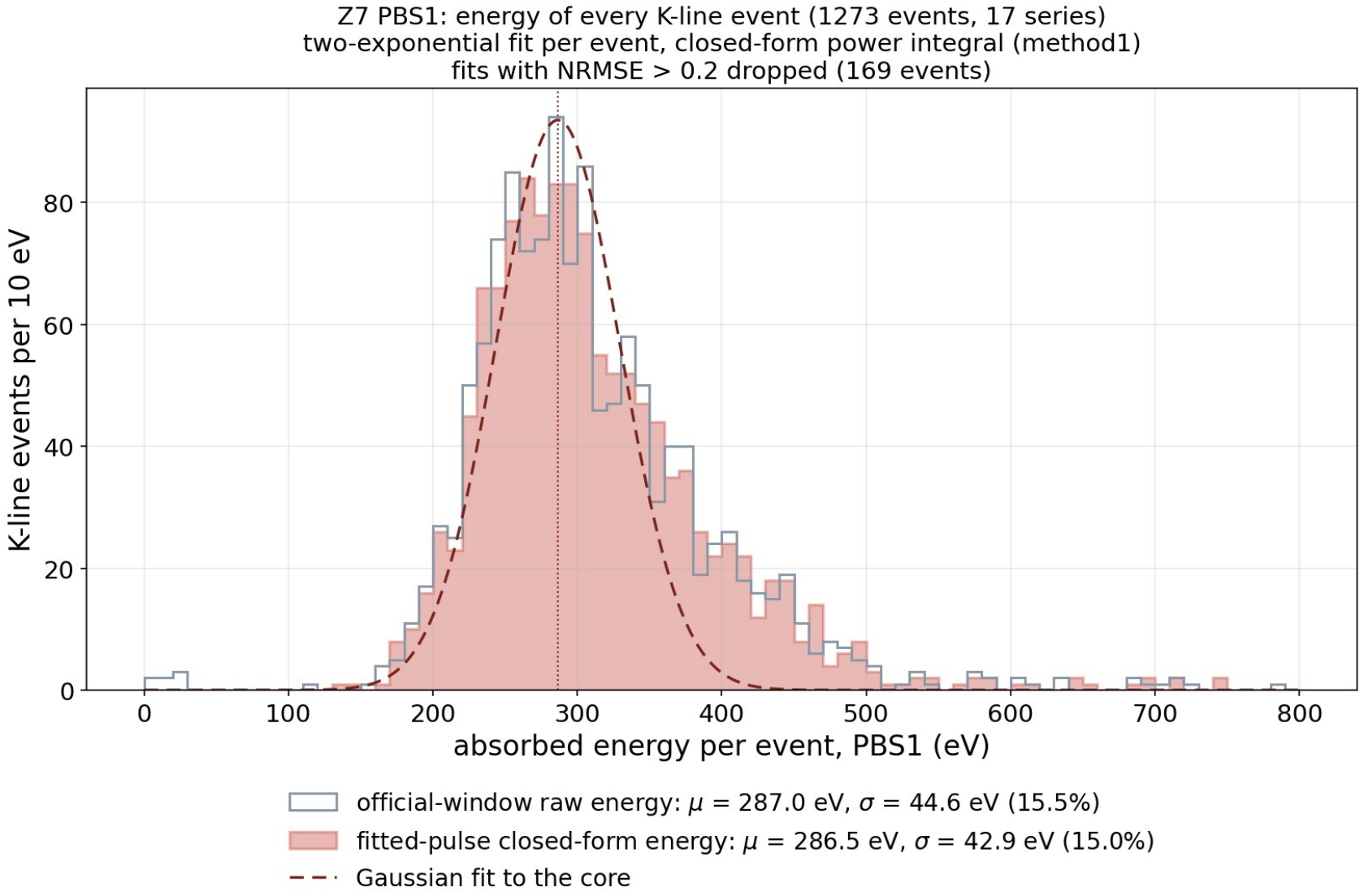
PFS2 is not read out on Z7, so side 2 has 5 channels
the channels also differ from each other; see the backup

The channels do not share the energy equally: 60 % of it lands on side 1, and PES1 alone has 489 eV against 195 eV in PES2.

Summed over the eleven channels: 3419 eV = 33.0 % of 10.37 keV. PDS2 (bottom right) carries a slow swing.

Now every event: one channel gives a wide, lopsided distribution

PBS1, 1273 K-line events, one fit and one integral each



Red: from the fit. Grey: the official window on the raw trace. Same peak to 0.2 %.

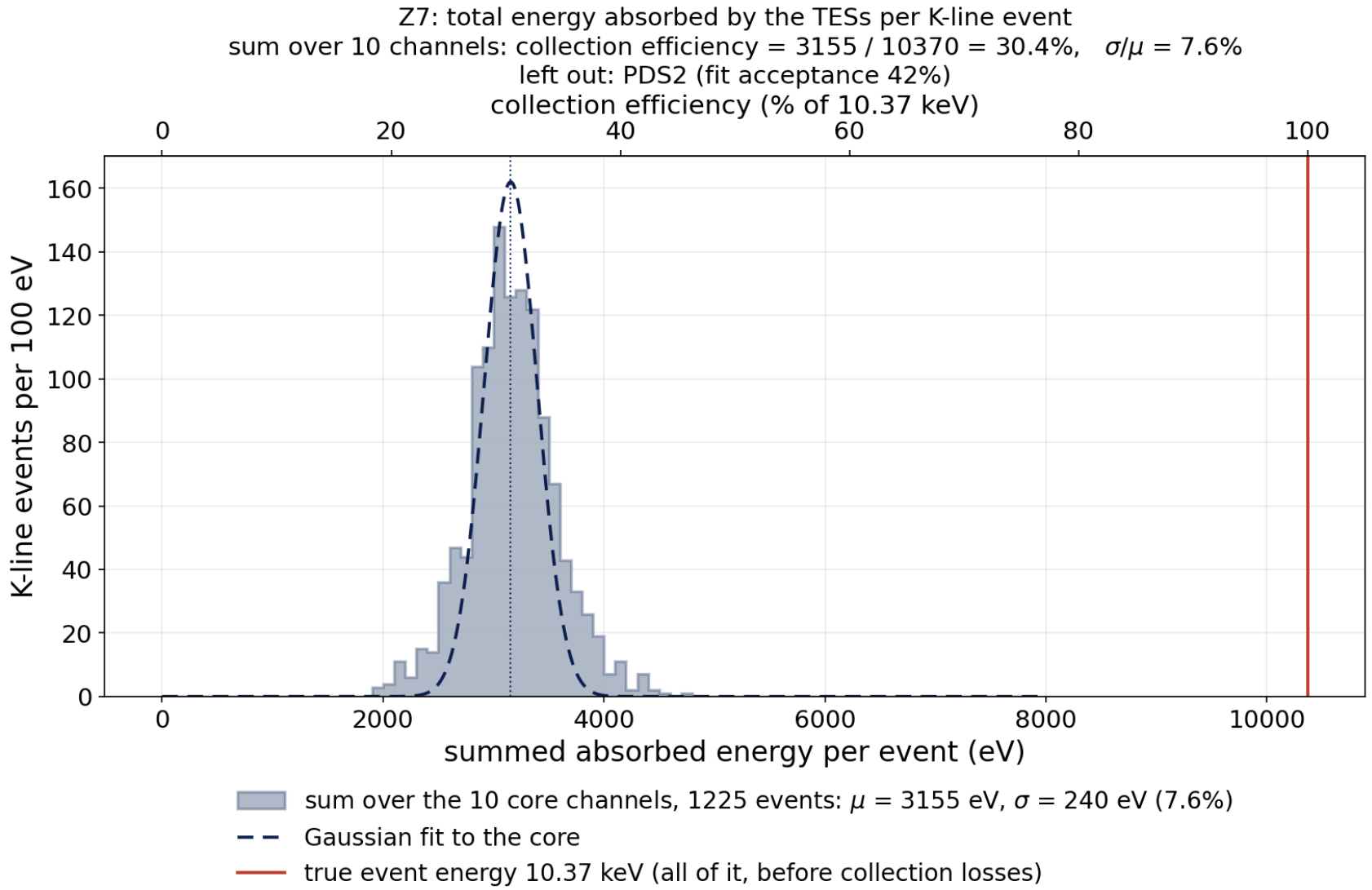
Peak 287 eV, width 15 % — for a line whose energy is fixed.

The tail to the right is the events that happened near this channel.

One channel's width is event position, not noise.

Summed over the channels: 30.4 %

the total energy the TESs absorbed per K-line event



Ten channels summed, 1225 events: 3155 eV = 30.4 %, and the width drops to 7.6 % — the position effect cancels.

PDS2 is left out: only 42 % of its events can be fitted, and those are the events that happened near it.

Collection efficiency:

30.4 % ± 2.3 % (σ/μ = 7.6 %).

Red line: 10.37 keV — two thirds of the energy never reaches the TESs.

Summary, and what is still open

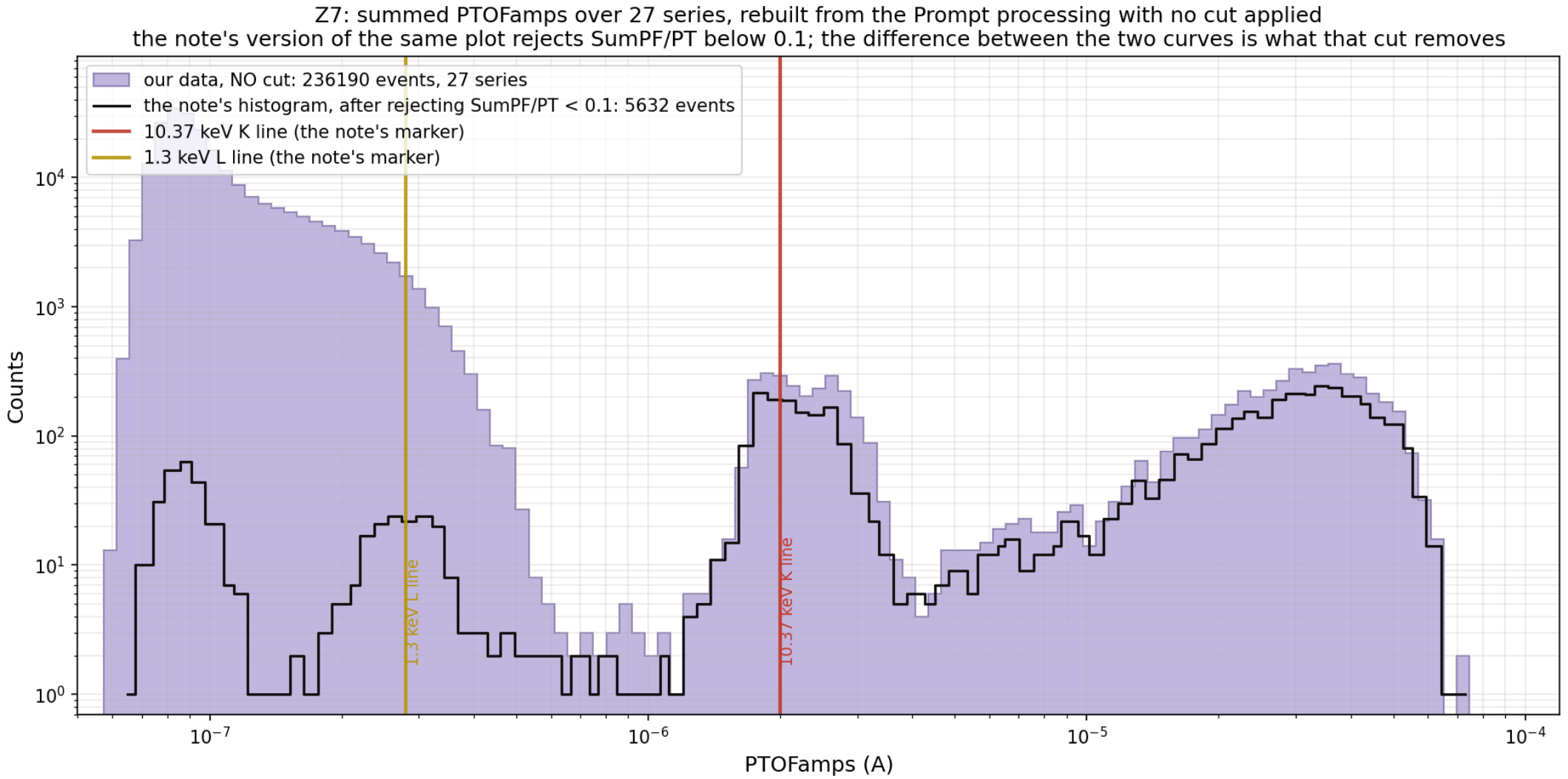
The current pulse becomes a power pulse through the TES equations; the area under the power is the energy.

The energy is taken from the fitted pulse, so the noise at the peak and the baseline drift do not enter.

Z7 absorbs 30.4 % ± 2.3 % (σ/μ = 7.6 %) of a 10.37 keV event in its TESs, summed over 10 channels. The CDMS note reports 26–41 % on the R37 CUTE tower.

Open: Z7 only · the bias of Z7 is not established (0 V assumed) · no dI/dV · PDS2 not included · 11 % of events fit in no channel · 13 later series (different bias point) not used.

Backup — the events



Backup — the two Methods of the CDMS note

$\delta P = I_0(R_L - R_0)\cdot\delta I + c_2\cdot R_L\cdot(\delta I)^2$ with $c_2 = 2$ (Method 1, used here) and $c_2 = -1$ (Method 2)

The two differ only in the quadratic term: 1.7 % in energy (PBS1, median of 15 events) — far below the 15 % single-channel width and the 7.6 % width of the sum.

The inductor term of the full equations needs L and the loop gain from dI/dV, which do not exist for these series on MSI; it is neglected.

The official Eabs (trigger bin 16383, baseline bins 93..15758, 5-pole 20 kHz prefilter, window −0.5/+1 ms, same formula) is reproduced to 0.2 %; the grey outlines on the histogram slides are it.

Backup — how the formulas are derived

1. From the current to the absorbed power

heat balance of the TES

$$C\dot{T} = P(t) + P_J - P_{bath}$$

over the whole pulse $\int C\dot{T}\,dt = 0$; the change in bath heat flow is not included

$$E = \int P\,dt = -\int \delta P_J\,dt$$

the circuit; at the bias point $V = I_0(R_L + R_0)$

$$V_{TES} = V - IR_L - L\dot{I}, \quad P_J = I V_{TES}$$

put $I = I_0 + \delta I$ and subtract the value at the bias point

$$\delta P_J = V\,\delta I - R_L\,(2I_0\,\delta I + (\delta I)^2) - L\,I\,\dot{\delta I}$$

integrate: the L term gives $L\,[I_0\,\delta I + (\delta I)^2/2]$, which is 0 at both ends

$$E = \int \delta P\,dt, \quad \delta P = I_0(R_L - R_0)\,\delta I + c_2\,R_L\,(\delta I)^2$$

c_2 = 1 from this derivation, 2 in Method 1 (used here), −1 in Method 2

2. Energy of the fitted pulse

the fit, with t counted from the pulse start t_0

$$\delta I(t) = A\,[\,e^{-t/\tau_f} - e^{-t/\tau_r}\,]$$

the only integral needed

$$\int_0^\infty e^{-t/\tau}\,dt = \tau$$

linear term

$$\int \delta I\,dt = A\,(\tau_f - \tau_r)$$

quadratic term: expand the square, integrate each exponential

$$\int (\delta I)^2 dt = A^2\left[\frac{\tau_f + \tau_r}{2} - \frac{2\tau_f\tau_r}{\tau_f + \tau_r}\right]$$

so, with $c_2 = 2$

$$E = I_0(R_L - R_0)\,A(\tau_f - \tau_r) + 2R_L\,A^2\left[\frac{\tau_f + \tau_r}{2} - \frac{2\tau_f\tau_r}{\tau_f + \tau_r}\right]$$

PBS1, event 30646: A = 1.39 μA, τ_r = 136 μs, τ_f = 213 μs ⇒ E = 305.2 eV

3. Symbols (Z7 PBS1 values) and assumptions

I_b: bias current R_{sh}: shunt resistor, 5.00 mΩ $V = I_b R_{sh}$: bias voltage, −1.146 μV R_p: parasitic resistance, 14.24 mΩ

$R_L = R_p + R_{sh}$: load resistance, 19.24 mΩ R_0: TES resistance at the bias point, 44.16 mΩ $I_0 = V/(R_L + R_0)$: TES current, −18.07 μA

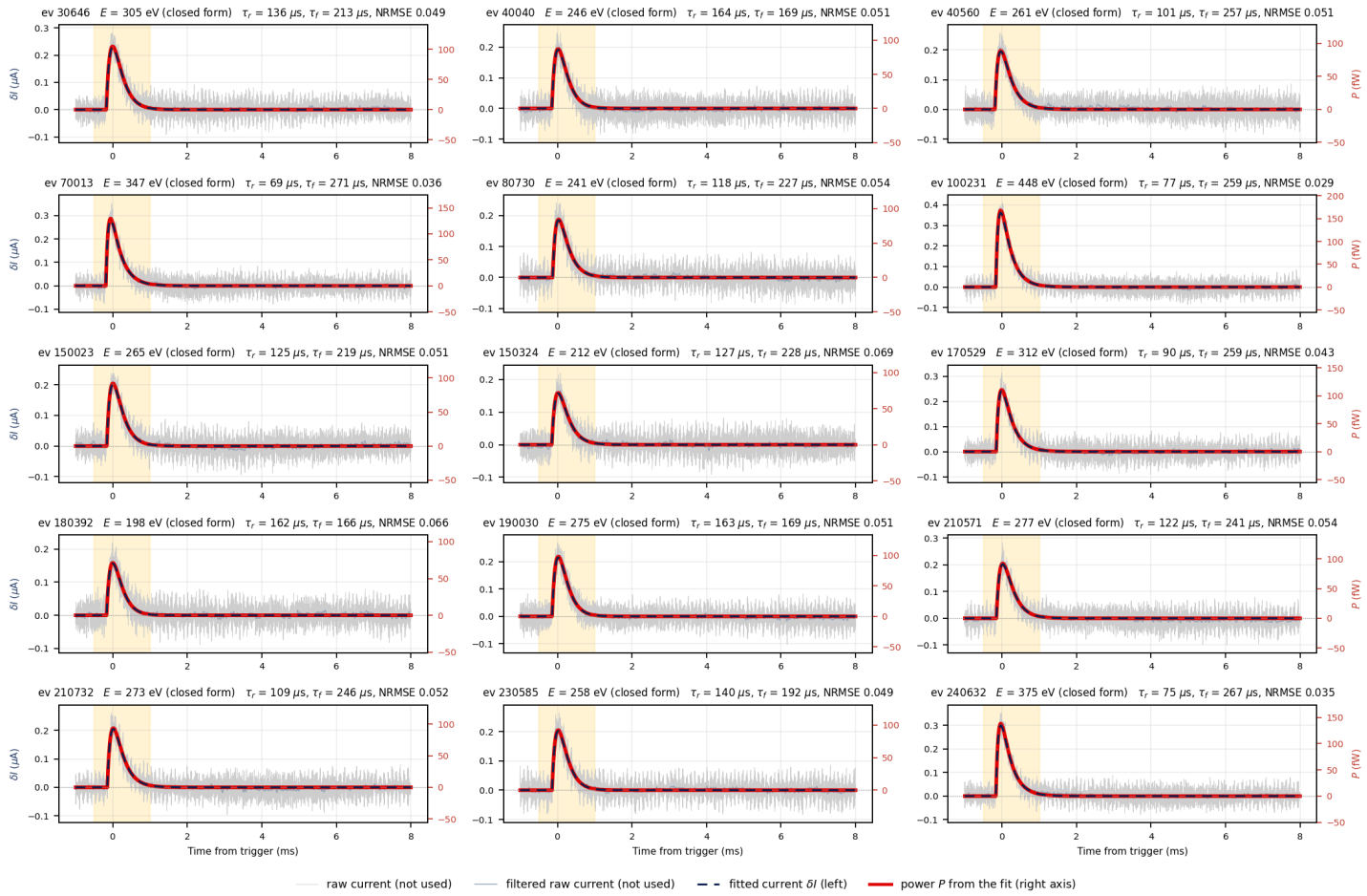
L: inductance (unknown) δI: current change A: amplitude τ_r, τ_f: rise and fall time constants t_0: pulse start

Assumptions: detector at 0 V (no NTL effect); the heat flow to the bath is not included;

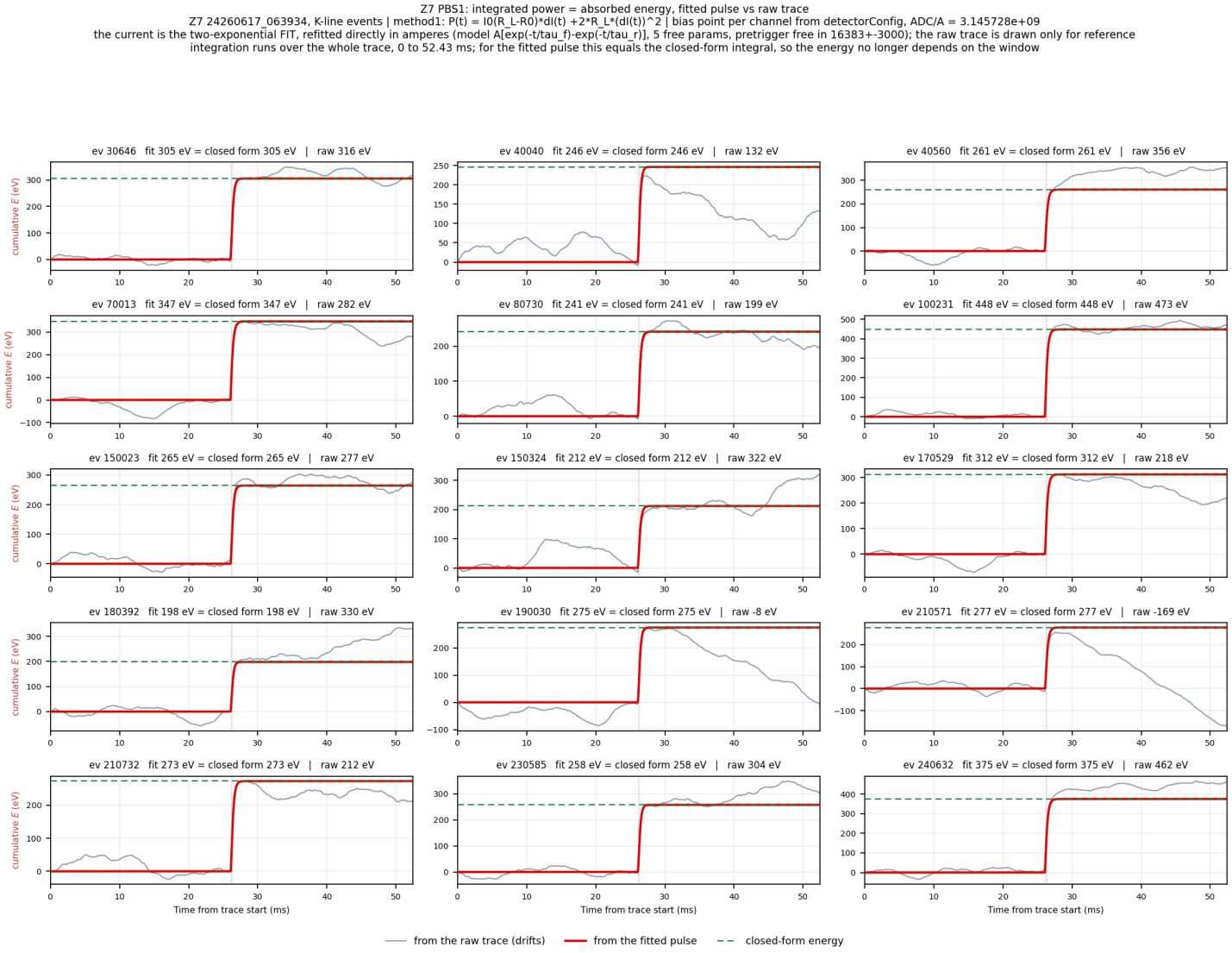
the L term cancels only after integration, so $\delta P(t)$ is not the exact power at each moment; only its integral is used

Backup — 15 events, fitted current and power (PBS1)

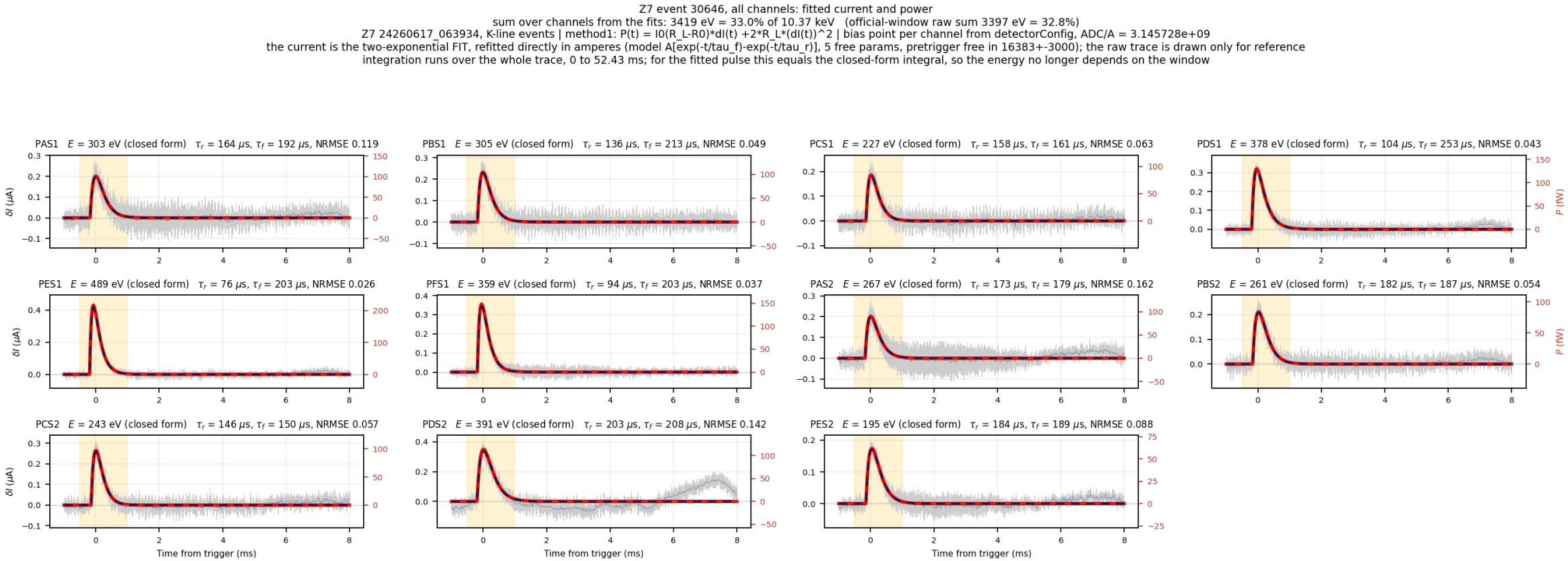
Z7 PBS1: fitted current pulse (dashed navy) and the power pulse it implies (red)
Z7 24260617_063934, K-line events | method1: P(t) = I0(R_L-R0)*dI(t) +2*R_L*(dI(t))^2 | bias point per channel from detectorConfig, ADC/A = 3.145728e+09
the current is the two-exponential FIT, refitted directly in amperes (model A[exp(-t/tau_f)-exp(-t/tau_r)], 5 free params, pretrigger free in 16383+-3000); the raw trace is drawn only for reference
integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



Backup — 15 events, cumulative energy (PBS1)

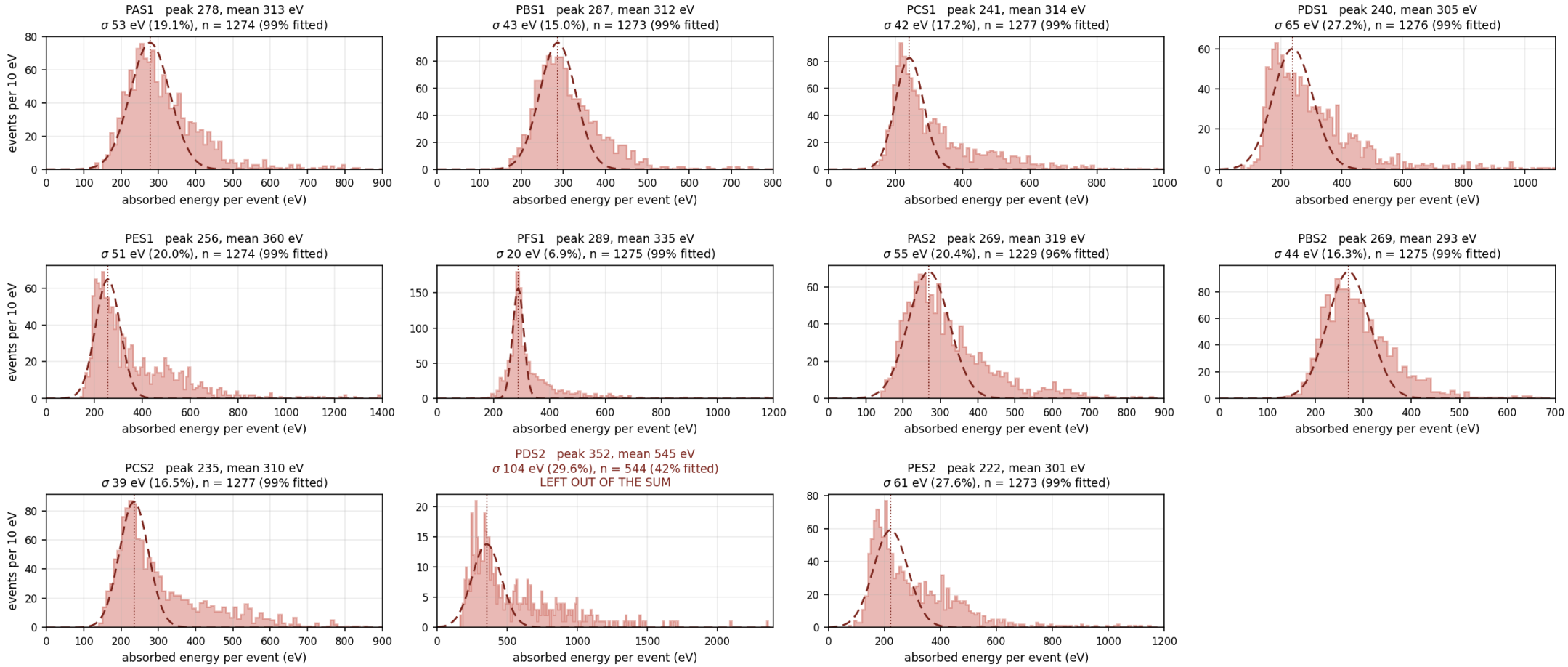


Backup — event 30646, all channels



Backup — energy distribution of every channel

Z7: energy of every K-line event, one panel per channel (17 series)
each entry is one event: its trace fitted with the two-exponential and the power integrated in closed form, method1 quadratic
the dashed curve is a Gaussian fit to the core, so its centre is the PEAK; these distributions are right skewed, so peaks must not be added (the plain means do add, to 3705 eV = 35.7% of the 10.37 keV K line)



Backup — event bookkeeping: what is excluded at each step

17 series, June 17–20 (the 13 series from June 21 on sit at a different bias point and are left out)

PTOFamps window around the K line (no trigger cut): 1442 events

− 158 events (11 %): the two-exponential fit fails in every channel (all NRMSE > 0.2) — pileup / glitches, not checked one by one

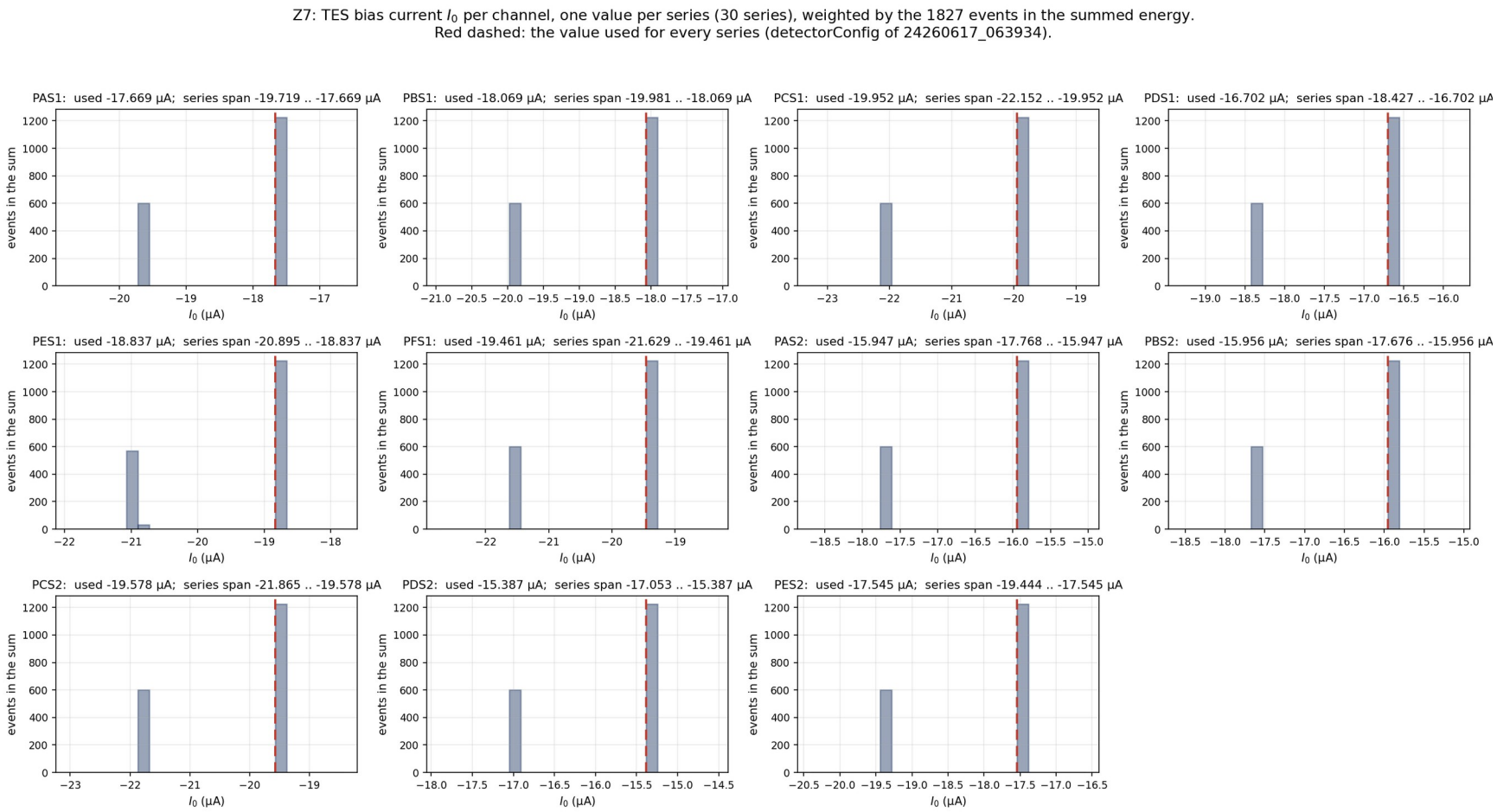
= 1284 events with at least one usable fit

− 59 events: at least one of the 10 core channels failed to fit — an incomplete sum would read low

= 1225 events (85 % of 1442) enter the summed-energy distribution

Channel-level cuts, applied separately: PDS2 left out (fits only 42 % of events, the ones near it) · PFS2 has no data on Z7

Backup — the TES bias current I_0 behind the coefficients



One I_0 per channel per series (detectorConfig, all 30 series), weighted by the events in the sum. Red dashed: the value used for every series (first series, e.g. PBS1 −18.07 µA).

Two bias states: June 17–20 (1225 events) at the values used; June 21–23 (602 events) with $|I_0|$ 10 % larger and R_0 20 % smaller. The fitted pulses are the same in both (energy and fall time within 2 %); the group's standard cdmsbats processing (Default_tag, per-series detectorConfig) gives 16.5 % less Eabs in the second state. Decision (Prof. Liu): the 13 later series are left out; the result uses the 17 series of the first state.